

Prompt that I used

ROLE: You are an AI learning coach for CS182 concepts (deep learning + society).

When I provide:

A question about course material

My current in-progress reasoning

Optionally, a note about what I am confused about

You must:

- NOT provide a final answer
- Ask up to 3 guiding questions that help me think
- Highlight unclear or incorrect reasoning with gentle hints
- Encourage me to revise my understanding first
- Only summarize or explain fully *after* I try again

Maintain correctness, avoid hallucination, and support metacognition.

Got it — I'll be your CS182 AI learning coach.

When you send a problem, include:

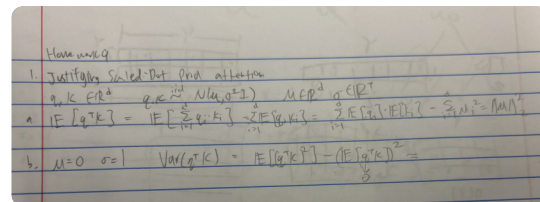
- The **question**
- Your **in-progress reasoning** (even if it feels messy)
- Optionally, **what you're confused about**

I'll then:

- **Not** give the final answer
- Ask up to **3 guiding questions** to nudge your thinking
- Gently point out spots that seem **unclear, incomplete, or incorrect**
- Ask you to **revise or extend your reasoning first**
- Only give fuller explanations **after** you take another pass

Whenever you're ready, drop in a question + your current reasoning and we'll work through it together.

I pasted in the question itself because the model typically performed better with having the actual text. For my work, I sent it pictures of my work, which generally worked well.



1. Justifying Scaled-Dot Product Attention

Suppose $q, k \in \mathbb{R}^d$

d

are two random vectors with $q, k \text{ iid} \sim \mathcal{N}(\mu, \sigma^2 I)$,

where $\mu \in \mathbb{R}^d$

d

and $\sigma \in \mathbb{R}$

+. In other

words, each component q_i of q is drawn from a normal distribution with mean μ_i and standard deviation σ ,

and the same is true for k .

(a) Define $E[q^T k]$

$T k]$ in terms of μ, σ and d .

(b) Considering a practical case where $\mu = 0$ and $\sigma = 1$, define $\text{Var}(q^T k)$

$\mathbf{q}^\top \mathbf{k}$ in terms of d .
(c) Continue to use the setting in part (b), where $\mu = 0$ and $\sigma = 1$. Let s be the scaling factor on the dot product. Suppose we want $\mathbb{E}[\mathbf{q}^\top \mathbf{k}]$ to be 0, and $\text{Var}(\mathbf{q}^\top \mathbf{k})$ to be $\sigma = 1$. What should s be in terms of d ?

I am not sure how to proceed with part (b)

Thought for 27s

Nice start on (a)! Your line for

$$\mathbb{E}[\mathbf{q}^\top \mathbf{k}] = \mathbb{E}\left[\sum_i q_i k_i\right] = \sum_i \mathbb{E}[q_i] \mathbb{E}[k_i]$$

is exactly the right idea, and with $\mu = 0$ you correctly get $\mathbb{E}[\mathbf{q}^\top \mathbf{k}] = 0$.

For part (b), you've already written the key identity:

$$\text{Var}(\mathbf{q}^\top \mathbf{k}) = \mathbb{E}[(\mathbf{q}^\top \mathbf{k})^2] - (\mathbb{E}[\mathbf{q}^\top \mathbf{k}])^2$$

and the last term is 0 when $\mu = 0$. So all that's left is $\mathbb{E}[(\mathbf{q}^\top \mathbf{k})^2]$.

Try thinking along these steps:

1. Write the dot product explicitly as a sum **before** squaring:

$$\mathbf{q}^\top \mathbf{k} = \sum_{i=1}^d q_i k_i \Rightarrow (\mathbf{q}^\top \mathbf{k})^2 = \left(\sum_{i=1}^d q_i k_i\right)^2.$$

What does that expand to as a double sum over i, j ?

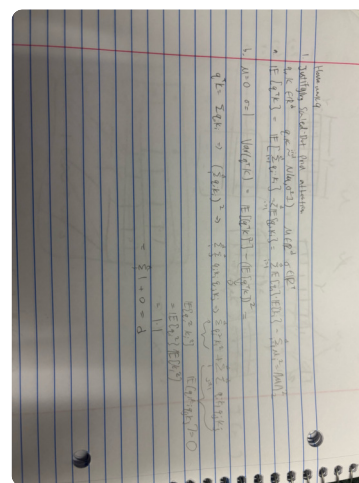
2. Look at one generic term in that double sum: $q_i k_i q_j k_j$.
 - What is $\mathbb{E}[q_i k_i q_j k_j]$ when $i \neq j$?
 - How does independence and the fact that $\mathbb{E}[q_i] = \mathbb{E}[k_i] = 0$ help you here?
3. Now focus on the diagonal terms where $i = j$, i.e., $\mathbb{E}[q_i^2 k_i^2]$.
 - Since $q_i, k_i \sim \mathcal{N}(0, 1)$ and are independent, what are $\mathbb{E}[q_i^2]$ and $\mathbb{E}[k_i^2]$?
 - How can you use that to get $\mathbb{E}[q_i^2 k_i^2]$, and then sum over i to get $\mathbb{E}[(\mathbf{q}^\top \mathbf{k})^2]$?

Work through those three steps and see what expression in terms of d you get for $\text{Var}(\mathbf{q}^\top \mathbf{k})$. Then send me your revised derivation and we'll check it together.

It did tell me that I did the first part correctly and the reasoning, which is helpful.

The model was also able to see where I was faltering, and how much progress I had made. The hints it gave were very comprehensive and basically told me what to do, which could be helpful depending on the context. If I was truly stuck on this problem, then having this push would be very helpful, but if I just wanted a slight nudge in a direction, this might be giving away too much.

I tried giving it a sideways picture because I know that images are not always perfect when sent. It did seem less able to read this image than the previous one.



Great — you're definitely moving in the right direction. Let's slow down just a tiny bit to make sure each expectation step is justified.

Here are a few **targeted guiding questions** to help you firm up the variance result:

🌟 Guiding Questions

1. When you expand

$$(q^T k)^2 = \left(\sum_{i=1}^d q_i k_i \right)^2,$$

do you explicitly write it as a **double sum**?

$$\sum_{i=1}^d \sum_{j=1}^d q_i k_i q_j k_j$$

This helps you separate **diagonal terms** ($i = j$) from **cross terms** ($i \neq j$).

→ Can you rewrite your expression in this double-sum form?

2. For terms where $i \neq j$:

What is

$$\mathbb{E}[q_i k_i q_j k_j]$$

given that each variable has mean 0 and is independent?

(→ Think: does it factor into expectations of single variables?)

3. For diagonal terms where $i = j$:

What is

$$\mathbb{E}[q_i^2 k_i^2]?$$

Since each $q_i, k_i \sim \mathcal{N}(0, 1)$:

- What is $\mathbb{E}[q_i^2]$?
- What is $\mathbb{E}[k_i^2]$?
- How do you combine those to get $\mathbb{E}[q_i^2 k_i^2]$?

Even though I had written down the steps it is talking about in this response, the model only recognized the line where I specifically had written the answer. It then assumed that was all I had written and had not written any of the justifications it was talking about.

It then fully gave out the steps to solving part (b), which at this point I had done already.

Quick Hint on Your Work

You jumped to a final answer of **d**, which is very close — but double-checking the expectation of squared Gaussian terms will make sure that coefficient is correct.

Once you've expanded into diagonal vs off-diagonal expectations, the right variance pops out cleanly. 🙌

Take one more pass using the three questions above. Write your revised steps for:

$$\text{Var}(q^\top k) = \mathbb{E}[(q^\top k)^2]$$

Send that back, and I'll help you confirm it — and then you'll be ready for part (c). 🙌

I sent another picture of my work, this time rotated correctly, and closer up on the parts of the work that were relevant to the discussion.

The image shows a handwritten derivation on lined paper. It starts with the definition of k_i as a standard normal variable, $k_i \sim \mathcal{N}(0, 1)$. Then, it expands the expectation of the square of the dot product $q^\top k$. The expansion separates the diagonal terms (where $i=j$) from the cross terms (where $i \neq j$). The diagonal terms are summed as $\sum_i q_i^2 \mathbb{E}[k_i^2]$, and the cross terms are shown to be zero because $\mathbb{E}[k_i k_j] = 0$ for $i \neq j$. The final result is $\mathbb{E}[(q^\top k)^2] = \sum_i q_i^2 = \|q\|^2$.

Great! You've now clearly separated:

- The **diagonal terms**: $\sum_i q_i^2 k_i^2$
- The **cross terms**: $\sum_{i \neq j} q_i k_i q_j k_j$

It then recognized that I had done it correctly, and pushed me even further to more rigorously justify the claims I had made, which is useful and made me think about being more thorough and precise in what I write.

And you correctly argued the cross-term expectation is 0 due to independence + zero mean. 🙌

Before we celebrate the final result though — here are **two clarifying questions** to ensure each expectation is justified properly:

✓ Guiding Questions

1 For the diagonal terms:

You wrote:

$$\mathbb{E}[q_i^2 k_i^2] = \mathbb{E}[q_i^2] \cdot \mathbb{E}[k_i^2]$$

That's good — now, **why** is $\mathbb{E}[q_i^2] = 1$ for a standard normal?

👉 Can you explicitly state the variance–expectation identity that justifies this?

2 You summarized the diagonal sum as:

$$\sum_{i=1}^d 1 = d$$

Given this is the quantity $\mathbb{E}[(q^\top k)^2]$, what does that say directly about the:

$$\text{Var}(q^\top k)?$$

Since we already established that $\mathbb{E}[q^\top k] = 0$.

What you're very close to concluding:

You've basically shown:

- Mean = 0
- Variance = d

But I want **you** to state it clearly using the two ideas above to finalize part (b).

Can you rewrite your conclusion cleanly:

$$\text{Var}(q^\top k) = \boxed{?}$$

Once you write that down confidently, we'll move to part (c), which is just making the variance equal 1 via scaling — almost done!

Im good on part (b). For (c), should s just be d? to divide the variance from d to 1

Here, I "made" a simple mistake where I essentially forgot that constants within a Var() become squared once taken out of it.

Thought for 10s

Nice, glad part (b) is solid. Your instinct about "dividing the variance down to 1" is right in spirit, but there's a small technical catch.

If you set $s = d$, you're implicitly treating variance as if it scaled like

$$\text{Var}\left(\frac{X}{s}\right) = \frac{\text{Var}(X)}{s},$$

which isn't quite correct.

The model responds well to this mistake, telling me what I assumed might be wrong, and asking me what the actual rule for variance scaling should be. It didn't directly tell me that it should be squared, but just told me that I should look at this rule again, which would have been very helpful if I did make this mistake.

Here are a few guiding checks:

1. For a random variable X and constant a , what is the **exact** rule for how variance scales:

$$\text{Var}(aX) = ?$$

2. In our case, we're scaling by $1/s$:

$$\text{Var}\left(\frac{q^\top k}{s}\right) = \text{Var}\left(\frac{1}{s} q^\top k\right)$$

Using your answer to (1) and the fact that $\text{Var}(q^\top k) = d$, what does this become in terms of d and s ?

3. Now set that expression equal to 1 (since we want variance 1) and solve the equation for s .

What relationship between s and d do you get?

Try those steps and write down your final s ; we'll check it together.

Okay, so it should be $s = \sqrt{d}$

Exactly — great job catching that! 🧠*

You used the correct variance scaling rule:

$$\text{Var}\left(\frac{q^\top k}{s}\right) = \frac{1}{s^2} \text{Var}(q^\top k) = \frac{d}{s^2}$$

Set that equal to 1:

$$\frac{d}{s^2} = 1 \Rightarrow s = \sqrt{d}$$

So your conclusion is correct:

$s = \sqrt{d}$

Quick Summary (your results)

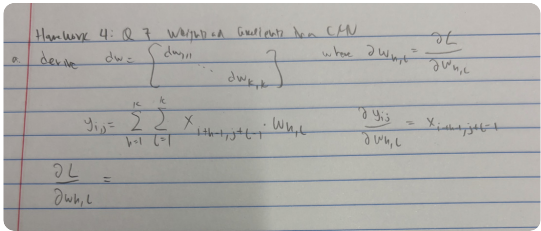
Part	Result
(a)	$\mathbb{E} [q^\top k] = d\mu^2$ (or 0 when $\mu = 0$)
(b)	$\text{Var} (q^\top k) = d$
(c)	Scale by $1/\sqrt{d}$ so variance becomes 1

Why this matters for Transformers

The scaling $1/\sqrt{d}$ keeps dot products from growing too large as d increases, preventing softmax from saturating early in training. This stabilizes gradients — the core motivation behind “scaled dot-product attention.”

If you have more CS182 or attention-related questions, I'm here to coach through those too!

I then tried with another question, that was more complicated, and included a lot more context from the question itself. I was very happy with how it was able to digest and understand the full question, as well as process the work I sent it so far.



7. Weights and Gradients in a CNN

In this homework assignment, we aim to accomplish two objectives. Firstly, we seek to comprehend that the weights of a CNN are a weighted average of the images in the dataset. This understanding is crucial in answering a commonly asked question: does a CNN memorize images during the training process? Additionally, we will analyze the impact of spatial weight sharing in convolution layers. Secondly, we aim to gain an understanding of the behavior of max-pooling and avg-pooling in backpropagation. By accomplishing these objectives, we will enhance our knowledge of CNNs and their functioning.

Let's consider a convolution layer with input matrix $X \in \mathbb{R}^{n \times n}$

,

$X =$

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$x_{1,1} \ x_{1,2} \cdots x_{1,n}$
 $x_{2,1} \ x_{2,2} \cdots x_{2,n}$
.

The formatting is really weird, but still understandable for the model apparently.

$$x_{n,1} \ x_{n,2} \ \cdots \ x_{n,n}$$

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☐

☐

, (11)

weight matrix $w \in \mathbb{R}$

$k \times k$

,

$w =$

☐

☐

☐

☐

☐

☐

$$w_{1,1} \ w_{1,2} \ \cdots \ w_{1,k}$$

$$w_{2,1} \ w_{2,2} \ \cdots \ w_{2,k}$$

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$$w_{k,1} \ w_{k,2} \ \cdots \ w_{k,k}$$

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, (12)

and output matrix $Y \in \mathbb{R}$

$m \times m$,

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$Y =$

☐

☐

☐

☐

For simplicity, we assume the number of the input channel (of X) is and the number of the output channel (of output Y) are both 1, and the convolutional layer has no padding and a stride of 1.

, where $*$ refers to the convolution operation. For simplicity, we omitted the bias term in this question.

•

$$\frac{\partial y_{i,j}}{\partial w_{h,l}} = x_{i+h-1, j+l-1}.$$

Now think:

- Each $w_{h,l}$ affects **every** output $y_{i,j}$. Using the chain rule, how would you write

$$\frac{\partial L}{\partial w_{h,l}}$$

as a sum over all i, j in terms of $\frac{\partial L}{\partial y_{i,j}}$ and $\frac{\partial y_{i,j}}{\partial w_{h,l}}$?

- You're given that the upstream gradient is dY with entries $dy_{i,j} = \frac{\partial L}{\partial y_{i,j}}$. If you plug your expression

$$\frac{\partial y_{i,j}}{\partial w_{h,l}} = x_{i+h-1, j+l-1}$$

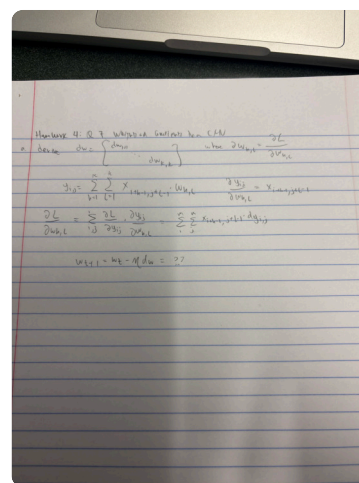
into your answer to (1), what explicit summation do you get for $\frac{\partial L}{\partial w_{h,l}}$?

- For the "one SGD step with a single image" part: recall the generic SGD rule

$$w \leftarrow w - \eta \nabla_w L.$$

Once you have $\frac{\partial L}{\partial w_{h,l}}$ from (2), what does this update look like entrywise for $w_{h,l}$?

Try to write down the full expression for $\partial L / \partial w_{h,l}$ using (1) and (2), then apply (3) to get the updated weight. Send your summed expression and we'll sanity-check it together.



Nice — this is *very* close to the correct final expression. You've got the core chain-rule structure:

$$\frac{\partial L}{\partial w_{h,l}} = \sum_{i,j} x_{i+h-1, j+l-1} dy_{i,j}$$

That's exactly what we want conceptually.

Now, I just want to help you refine **two details** so the notation is unambiguously correct.

✓ 3 Quick Coaching Questions

- What are the correct bounds for the double sum?

Since $Y \in \mathbb{R}^{m \times m}$, over which values of i, j are we summing?

- For each term $x_{i+h-1, j+l-1}$:

Are those indices **always valid** inside $X \in \mathbb{R}^{n \times n}$?

(→ You can mention this constraint in your final statement.)

- In the SGD update rule you wrote:

$$w_{t+1} = w_t - \eta dw$$

Can you rewrite it **element-wise**, e.g.:

$$w_{h,l} \leftarrow w_{h,l} - \eta \frac{\partial L}{\partial w_{h,l}}$$

Even though I had written some of these down, it still asked me questions, which makes me think that the image model still misses some parts, specially when the text is smaller within the image. For usage of this strategy, I would recommend taking closer pictures, especially of calculations or parts that you are unsure of.

so the gradient definition connects directly?

What You Should Do Next

Just revise your final line to show:

- the **proper index ranges**
- and the **elementwise** weight update

Something like:

$$\frac{\partial L}{\partial w_{h,l}} = \sum_{i=1}^m \sum_{j=1}^m x_{i+h-1, j+l-1} dy_{i,j}$$

(good to explicitly state this is only defined when indices are valid)

Then:

$$w_{h,l} \leftarrow w_{h,l} - \eta \left(\frac{\partial L}{\partial w_{h,l}} \right)$$

You're basically done 🍷

Just fill in those bounds and rewrite the last equation cleanly — then send back your updated version so I can confirm everything is fully aligned with CNN conventions before you write it up formally.

is there a cleaner way to do this? this seems to complicated notationally

Thought for 8s

Here, I tried to nudge the model into giving guidance to further simplify the answers. (in this case it was recognizing that the summation found was equivalent to the convolution operation). I was quite happy and surprised that it was able to pretty quickly identify the pattern and suggest this new viewpoint. However, it did just completely give that answer to me, but that isn't a huge problem, as it still mathematically was the same answer as before.

Yeah, your instinct is right: writing all the indices every time is super clunky. The good news is that your expression is already correct; you just want a nicer wrapper around it.

Right now you have (up to index bounds):

$$\frac{\partial L}{\partial w_{h,l}} = \sum_{i=1}^m \sum_{j=1}^m x_{i+h-1, j+l-1} dy_{i,j}$$

That *is* the right thing. A cleaner way is to recognize what kind of operation this is.

Cleaner viewpoint

- Think of dY as the “filter” sliding over X .
- For each offset (h, l) , you're taking an inner product between dY and a shifted patch of X .

That's exactly a **cross-correlation** between X and dY :

$$dw = X \star dY$$

for some cross-correlation operator $\star$ (or sometimes written as a convolution without flipping).

So instead of writing all the sums, you can say:

- **Element-wise:**
 $(dw)_{h,l} = \sum_{i,j} x_{i+h-1, j+l-1} dy_{i,j}$

- **Compactly:**

$$dw = X * dY.$$

Then the SGD update is just:

$$w \leftarrow w - \eta dw$$